

1. Let's get a sense of how large powers of two really are.
 - (a) What is 2^{10} exactly?
 - (b) Assume that $2^{10} = 10^3$, one thousand. Using exponent rules, what are 2^{20} and 2^{30} as powers of 10? What rule can you apply to estimate 2^{10k} (at least for small positive integers k)?
 - (c) Using the internet, fill in the blanks in the following table. For the last two lines, choose your own examples!

Power of 2	Interpretation
2^{66}	Number of cells in a human body
	Number of different 128-bit RSA keys (N)
	Number of atoms in the observable universe
	Largest-bit RSA key ever factored

2. Assume the time to break an RSA key of length n bits is proportional to $2^{n/2}$ and suppose a 128-bit RSA key takes 1 day to break.
 - (a) How long would it take to break a 256-bit key under this model?
 - (b) Suppose that, due to faster hardware, computing power doubles every two years. How long would it take to break a 256-bit key in 20 years?
 - (c) How many bits should you add to the key to maintain the same level of security in 20 years?
3. Now let's assume that the time to break an RSA key of length n bits is proportional to $e^{n^{1/3}}$ (this is a very rough approximation of the asymptotic runtime of the Generalized Number Field Sieve algorithm for factoring). You would again like to answer the question: if computing power doubles every two years, how many bits should you add to the key to maintain the same level of security in 20 years? You may simply evaluate $e^{n^{1/3}}$ for various values of n to get a sense of how the runtime grows with n and then extrapolate from there.
4. Prove the following:
 - (a) If $c|a$ then $c|sa$ for any integer s .
 - (b) If $c|a$ and $c|b$ then $c|a + b$.
 - (c) If $c|a$ and $c|b$ then $c|sa + tb$ for any integers s, t .
 - (d) If $b > 0$, then $\gcd(a, b) = \gcd(a, r)$ where r is the remainder when a is divided by b .